

1. DEFINITION

DEFINITION
Propylene glycol monocaprylate (type I) EP / Propylene Glycol Mono and Dicaprylate NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Propylene Glycol Caprylate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	QJL3I5WV4F
ADDITIONAL INFORMATION ON THE DENOMINATION USP/FDA Substance Registration System (SRS) - Synonyms	CAPRYOL PGMC OCTANOIC ACID, ESTER WITH 1,2-PROPANEDIOL PROPYLENE GLYCOL CAPRYLATES PROPYLENE GLYCOL MONO AND DICAPRYLATE PROPYLENE GLYCOL MONOCAPRYLATE TYPE I [NF] PROPYLENE GLYCOL OCTANOATES
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Propylene glycol and Caprylic acid (C8). No solvent or catalyst used during the manufacturing process.

COMPOSITION
Well defined multi-constituent substance composed of Propylene glycol mono- and di- esters of Caprylic acid (C8), the monoester fraction being predominant (Specification: 55-80%)



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COMPOSITION	
Full Labeling (Key letters system)	
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Propylene Glycol Caprylate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	18895
Canada MF (Type III – Excipient)	2008-193
For submission of the DMF in another country, please contact your local Gattefossé representative	

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Propylene Glycol Monocaprylate (type I) Monograph applicable July 1, 2023
USP/ NF Current edition	Propylene Glycol Mono and Dicaprylate NB: may be labelled Propylene Glycol Monocaprylate type I until Dec. 01, 2026



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6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓ (warning for cats: 21CFR 582.1666)		✓ (warning for cats: 21CFR 582.1666)	✓ (5) 21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids	✓ 40CFR§180.1250	✓ 40CFR§180.1250
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Propylene Glycol Monocaprylate	Not available as an Active Ingredient in any application. Available for use as an Excipient Ingredient in: Over the Counter, Prescription Medicines. Not available as an Equivalent Ingredient in any application.	Not authorized in listed medicines.
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			



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REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA [NHPID](#)

Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Propylene Glycol Caprylate	<u>Toxicity Restriction:</u> Upper Limit: 25.0 Milligrams/kilogram body weight/day Detail: Acceptable daily intake up to 25 mg/kg bw/day, expressed as propylene glycol equivalents. Reference: WHO_JECFA <u>Roles Non-medicinal:</u> Purposes: Emulsifying Agent , Emulsion Stabilizer , Skin-Conditioning Agent - Emollient , Surfactant - Emulsifying Agent <u>General:</u> Route Of Administration: Topical Detail: For topical use only for the purposes of skin-conditioning agent - emollient and surfactant - emulsifying agent, when formulated to be non-irritating. Reference: CIR Route Of Administration: Oral Detail: For oral use only for the purposes of emulsifying agent and emulsion stabilizer. Reference: FCC
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute irritation (OECD 404)	Dermal	Rabbit	CAPRYOL™ 90 pure Vehicle: none	Mildly Irritant	Gattefossé data, GAT -98415, 1998
Acute irritation (OECD 405)	Ocular	Rabbit	CAPRYOL™ 90 pure Vehicle: none	Moderately Irritant	Gattefossé data, GAT -98414, 1998
Acute toxicity (OECD 401)	Oral	Rat	CAPRYOL™ 90 pure Vehicle: none	LD50 _(oral) > 5003 mg/kg	Gattefossé data, GAT -98416, 1998
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	CAPRYOL™ 90 Vehicle: Physiological saline solution M: 25, 50, 100 and 75 mg/kg F: 25, 50, 100, 150 and 125 mg/kg	MTD _(IV) M : 75 mg/kg MTD _(IV) F : 100 mg/kg	Gattefossé data, GAT-MTDs-PH- 08/0472, 2009

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SAFETY STUDIES PERFORMED BY GATTEFOSSÉ

Study type/ duration	Route	Species	Test article	Results/Conclusions	Reference
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	CAPRYOL™ 90 Vehicle: Physiological saline solution M+F: 75 mg/kg	MTD _(IV) < 75 mg/kg Administration at 75 mg/kg induce mortality (1/5 female and 1/5 male).	Gattefossé data, GAT-TAIs-PH- 08/047, 2009
Subchronic 7-day toxicity study (DRF)	Oral (Gavage)	Rat	CAPRYOL™ 90 Vehicle: none Control: deionized water Dose levels: 0, 300, 1000 and 2500 mg/kg/day	MTD _(oral) > 2500 mg/kg/day NOAEL _(oral) : 2500 mg/kg/day	Gattefossé data, GAT-WIL-261013, 2006
Subchronic 28-day toxicity study	Oral (Gavage)	Rat	CAPRYOL™ 90 Vehicle: none Control: deionized water Dose levels: 0, 500, 1500 and 2500 mg/kg/day	NOAEL _(oral) > 2500 mg/kg/day	Gattefossé data, GAT-WIL-261014, 2006
Embryo-fetal development (Segment II)	Oral (Gavage)	Rat	CAPRYOL™ 90 Vehicle: none Control: deionized water Dose levels: 0, 500, 1500 and 2500 mg/kg/day	Maternal NOAEL _(oral) : 2500 mg/kg/day Embryo/foetal NOAEL _(oral) : 2500 mg/kg/day	Gattefossé data, GAT-WIL-261015, 2007)
Ames test (With and without activation) (OECD 471)	In vitro	S. typhimurium	CAPRYOL™ 90 pure	Negative. Not mutagenic.	Gattefossé data, GAT-99443, 1999

OECD : Organisation for Economic Co-operation and development - JORF: Journal officiel République Française (Official Journal French Republic)

Summary of files available upon request, please contact your Gattefossé local representative.
Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview"
for a complete review on regulatory, tox and safety aspects.

CIR COMPENDIUM (Cosmetic Ingredient Review) (<http://www.cir-safety.org>)

Denomination	Reference	Maximum "as used" concentration	Conclusions
Propylene glycol esters	January 15 th , 2015	Up to 50%	On the basis of the available information presented in this report, the panel concludes that Propylene glycol esters (Propylene glycol laurate...) are safe as cosmetic ingredients in the present practices of concentration and use.



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9. RSPO CERTIFICATION

ROUNDTABLE ON SUSTAINABLE PALM OIL http://www.rspo.org/certification	
Product status	Gattefossé RSPO certification number
Product certified RSPO Mass Balance. More information on: www.RSPO.org	Gattefossé RSPO certification number: BVC-RSP0-1-1106949778 - from 01 July 2023

10. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EINECS #	Chemical name
Propylene Glycol Caprylate	85883-73-4 (or 68332-79-6) (or 31565-12-5) (or 99811-66-2)	288-722-4 (or none) (or none) (or 309-056-3)	Fatty acids, C6-12, esters with propylene glycol (or Octanoic acid, ester with 1,2-propanediol) (or Octanoic acid, monoester with 1,2-propanediol) (or Fatty acids, C4-22, esters with propylene glycol)

NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Propylene Glycol Caprylate	Listed	NDSL listed	Listed	Listed	Fatty acids, C6-12, esters with propylene glycol: AIIC, EINECS, NZIoC, REACH, TCSI, VNECI Octanoic acid, ester with 1,2-propanediol: NDSL, TSCA, VNECI Octanoic acid, monoester with 1,2-propanediol: NZIoC, TCSI, VNECI Fatty acids, C4-22-esters with propylene glycol: IECSC, EINECS, REACH, VNECI / Fatty acids, (C4-22), esters with propylene glycol: AREC, ECL

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation.
"Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer.
Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.



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REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Propylene Glycol Caprylate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

11. HISTORY OF USE

11.1. Pharmaceutical applications

11.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
No FDA IID reference exactly matches the definition Propylene Glycol Mono and Dicaprylate				
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

11.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.



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FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Propylene Glycol Monocaprylate (UNII: RT9P9S09QI)	ORAL	Tablet, film coated		24 mg
Propylene Glycol Dicaprylate (UNII: 581437HWX2)	TOPICAL	Cream	10% w/w	
Propylene Glycol Monolaurate (UNII: 668Z5835Z3)	ORAL	Capsule		2115 mg
	ORAL	Tablet		20 mg
	ORAL	Tablet, film coated		20 mg
	TOPICAL	Gel	3% w/w	
	TRANSDERMAL	Film, extended release	NA	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

11.1.1. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	Functional Category	Regulatory status	Safety
Propylene glycol monocaprylate	Emollient, Emulsion stabilizing agent, nonionic surfactant, solubilizing agent	Included in the FDA Inactive Ingredient Database (tablets). Included in nonparenteral medicines licensed in the UK. Included in topical medicines in Germany. Included in the Substances Added to Food (formerly EAFUS) list compiled by the FDA (propylene glycol mono- and di- esters of fats and fatty acids). Included in the Canadian Natural Health Products Ingredients Database.	The Cosmetic Ingredient Review (CIR) Expert Panel have assessed propylene glycol monocaprylate and propylene glycol dicaprylate separately, and concluded that propylene glycol dicaprylate is safe as used in the present practices of use and concentration in cosmetic formulations when formulated to be nonirritating.(CIR) At the time of the review, propylene glycol monocaprylate was not in use, but was concluded to be safe for future use if used in cosmetic formulations at concentrations comparable to propylene glycol dicaprylate. Propylene glycol monocaprylate was found to be slightly eye/skin irritant in rabbits. In rats, no adverse effects were observed for oral administration at the highest dose tested; 2.003 g/kg/day.



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11.2. Food/Nutraceutical/Dietary supplement applications

Propylene glycol esters of Fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 477 Propane-1,2-diol esters of fatty acids
U.S.A.	DSHEA (Dietary Supplement Health and Education Act) "Grandfathered ingredient" authorized for use in Dietary supplements	Esters of propylene glycol
U.S.A.	FCC (Food Chemical Codex)	Propylene glycol mono- and diesters
U.S.A.	USFA US Food additives according to 21 CFR	21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids
JAPAN	JSFA (Japanese Standards of Food Additives)	Propylene glycol esters of Fatty acids

HANDBOOK of FOOD ADDITIVES

Denomination	Use	Safety
CAPRYOL PGMC Propylene Glycol Caprylate	Solubilizer, stabilizer for foods	FDA 21CFR §172.856; E 477; worldwide food additive status

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s') properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s') suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

